Theories of Computation

Revision lecture 2

Exercise 1

$E = \underbrace{(b|ab)^*}\,(a|\varepsilon)$

1) i) Yes

ε matches $(b|ab)^*$ as $\varepsilon = (b|ab)^0$

choose ε in 2nd part

ii) abba Yes

ab matches $(b|ab)$

b ______

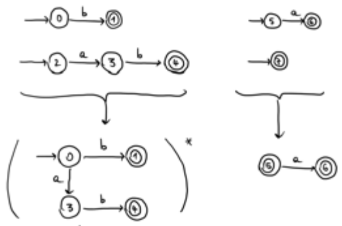
a matches $(a|\varepsilon)$

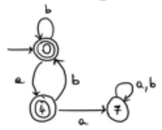
iii) aaa No

1st part cannot give a alone

2nd part can only give 1 a

2) $E = (b|ab)^*\,(a|\varepsilon)$





states

i. pairwise inequivalent

ii. reachable

(iii. hopeful)

i. 0 & 4 are inequivalent

because 0 accepts a and 4 does not

0 & 7 ______

because 0 accepts ε and 7 does not

4 & 7 ______

because 4 accepts ε and 7 does not

ii. 0 is reachable by ε

4 ______ a

7 ______ aa

Exercise 2 $\Rightarrow S ::= XX$

$X ::= aXa \mid bXb \mid a \mid b \mid \varepsilon$

(a)

Grammar $\mathcal{G}$ is ambiguous

because word bb has two distinct derivation trees in $\mathcal{G}$

$S \Rightarrow XX \Rightarrow bX \Rightarrow bb$

leftmost derivation

$S \Rightarrow XX \Rightarrow Xb \Rightarrow bb$

rightmost derivation

2 derivations for one word DOES NOT show ambiguity

(b) $X ::= \varepsilon$

$\Rightarrow S_0 ::= S$

$S ::= XX \mid \varepsilon \mid X$

$X ::= AU \mid BV \mid a \mid b$

$U ::= XA \mid A$

$V ::= XB \mid B$

$A ::= a$

$B ::= b$

Exercise 3

(a)

Tape content + head position	State	Instruction
… ⊔⊔aa⊔⊔ …	0	Read a

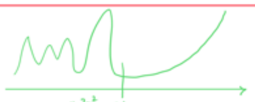
... ⊔⊔aa⊔⊔ ...	1	Write ⊔
... ⊔⊔⊔a⊔⊔ ...	2	Right
... ⊔⊔⊔a⊔⊔ ...	3	Read a
... ⊔⊔⊔a⊔⊔ ...	R	Right
... ⊔⊔⊔a⊔⊔ ...	3	Read ⊔
... ⊔⊔⊔a⊔⊔ ...	4	Left
... ⊔⊔⊔a⊔⊔ ...	5	Read a
... ⊔⊔⊔a⊔⊔ ...	6	Write ⊔
... ⊔⊔⊔⊔⊔⊔ ...	7	Left
... ⊔⊔⊔⊔⊔⊔ ...	8	Read ⊔
... ⊔⊔⊔⊔⊔⊔ ...	9	Right
... ⊔⊔⊔⊔⊔⊔ ...	0	Read ⊔
... ⊔⊔⊔⊔⊔⊔ ...	T	Return True

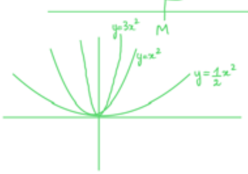
$n > 0$ even $\quad n = 2p+2 \quad p \geq 0$
$n > 0$ odd $\quad n = 2q+1 \quad q \geq 0$

(ii) Let $T(n)$ be the running time of $\mathcal{M}$ on an input of length n.

$T(n) \in O(n^2)$ — $g(n)$

→ there exists two numbers M and C such that : for all $n \geq M$, $T(n) \leq C.n^2$ — $g(n)$





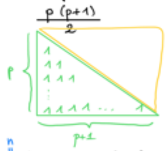
$$T(n) = \begin{cases} T_{even}(n) & \text{when } n \text{ even} \\ T_{odd}(n) & \text{when } n \text{ odd} \end{cases}$$

$$T_{even}(n) = T_{even}(2p+2) = \left(\sum_{k=0}^{p}(8k+12)\right) + 2$$

$$= 8\left(\sum_{k=0}^{p} k\right) + 12\left(\sum_{k=0}^{p} 1\right) + 2$$

$(8\times0+12\times1) + (8\times1+12\times1) + (8\times2+12\times1) + \dots + (8\times p+12\times1)$

$$= 8\times\underbrace{\boxed{(0+1+2+\dots+p)}}_{\frac{p(p+1)}{2}} + 12\times\underbrace{\boxed{(1+1+1\dots+1)}}_{p+1}$$



$$T_{even}(\overset{n}{2p+2}) = 8\,\frac{p(p+1)}{2} + 12(p+1) + 2$$

$$= 4p^2 + 4p + 12p + 12 + 2$$

$p = \frac{n-2}{2}$

$$= 4p^2 + 16p + 14$$

$$= 4\left(\frac{n-2}{2}\right)^2 + 16\,\frac{n-2}{2} + 14$$

$$= n^2 - 4n + 4 + 8n - 16 + 14$$

$$\boxed{T_{even}(n) = n^2 + 4n + 2}$$

for $n \geq 1$ (M). $T_{even}(n) \leq n^2 + 4n^2 + 2n^2 \leq 7n^2$
$\leq 12n^2$ (C)

$\begin{cases} n \leq n^2 \\ 1 \leq n^2 \end{cases}$

Hence $T_{even}(n) \in O(n^2)$

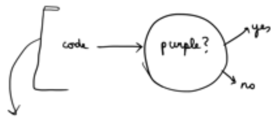
Exercise 4

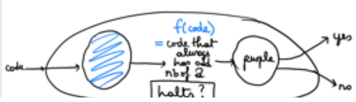
property of being purple is

$\boxed{\text{either halts or contains an even nb of 2's}}$

Suppose purple-ness is decidable.
$\boxed{\text{Want to show}}$ that this would imply halting to be decidable.
This would contradict the Halting Theorem

$$\frac{\begin{matrix} A \\ \vdots \\ \bot \end{matrix}}{\neg A}$$





$$f(\text{code}) = \begin{cases} \text{code} & \text{if has odd nb of a's} \\ \text{code} \\ \text{/*a/*} & \text{if code had even nb of a's} \end{cases}$$

code halts if and only if f(code) is purple

halting is decidable $\longleftarrow$ purple is decidable